RATING METHODOLOGY

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Rating Methodology

Oilfield Services

This rating methodology replaces the *Oilfield Services* methodology published on 25 January 2023. While this methodology reflects many of the same core principles as the 2023 methodology, we have made changes to the scorecard to provide greater consistency of approach across our corporate sector methodologies. In the Scale factor, we have replaced the EBITDA subfactor with Revenue. We have also made editorial changes throughout the methodology to enhance readability, including expanding the "Other considerations" section for enhanced consistency across our corporate sector methodologies, and we have changed the presentation of the scorecard.

Scope

This rating methodology applies to companies globally that are primarily engaged in providing oilfield services. Oilfield services companies are a diverse group of companies that provide products and services used in the exploration and production of oil and natural gas, including contract drilling companies, offshore support companies, seismic data providers and companies that manufacture equipment used by oilfield services companies.

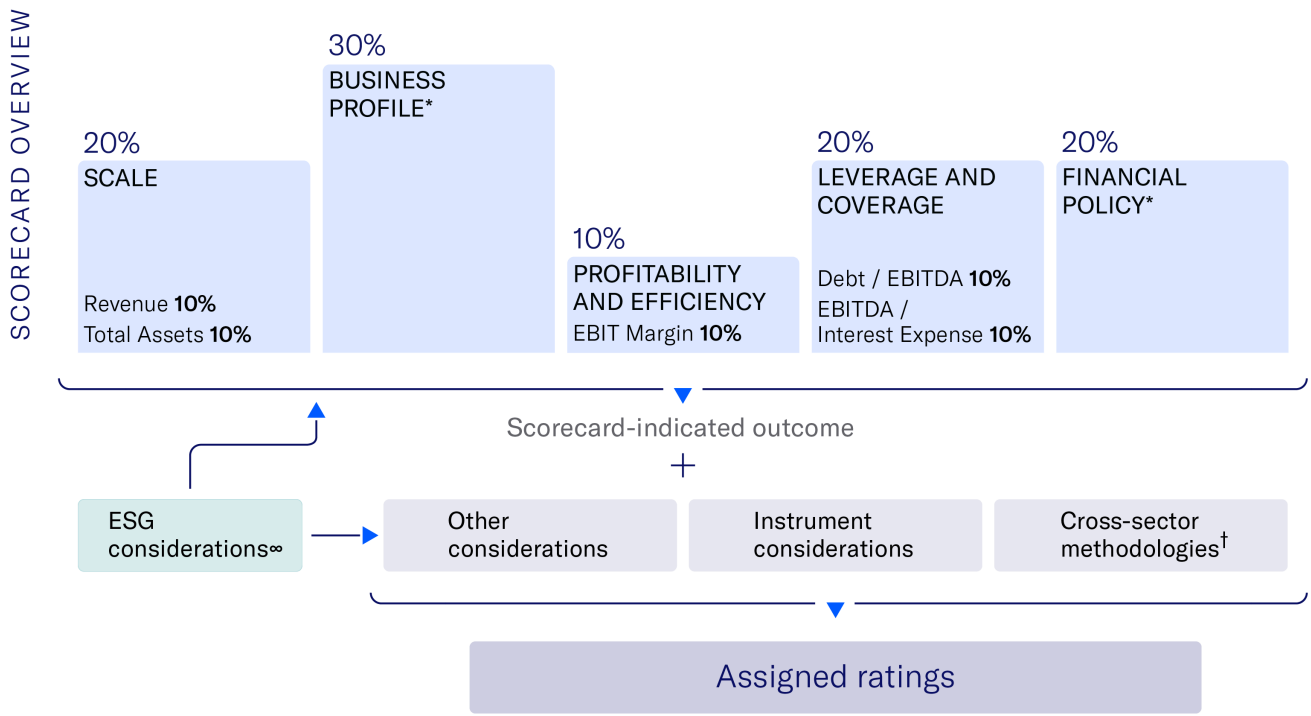
The determination of a company's primary business is generally based on the preponderance of the company's business risks, which are usually proportionate to the company's revenue, earnings and cash flows.

Rating approach

In this methodology, we explain our general approach to assessing credit risks of companies in this sector, including the quantitative and qualitative factors likely to affect rating outcomes.

The following exhibit illustrates our overall rating approach in the analysis of companies in this sector, which includes the use of a scorecard. The scorecard-indicated outcome may not match a company's assigned rating, which is determined by a rating committee. For more information, see the "Other considerations" and "Limitations" sections.

Exhibit 1
Illustration of the oilfield services rating approach



* This factor has no subfactors.
∞ Environmental, social and governance (ESG) considerations, including, where available, our opinions of exposure to them as expressed in Issuer Profile Scores (IPSs), may affect scorecard factors and other considerations outside of the scorecard. For more information, see the "Other considerations" section.
† Some of the methodological considerations described in one or more cross-sector methodologies may be relevant to ratings in this sector. A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.
Source: Moody's Ratings

Scorecard

The scorecard in this methodology is a relatively simple reference tool that can be used in most cases to approximate credit profiles of companies in this sector and to explain, in summary form, many of the factors that are generally most important in assigning ratings. The scorecard may not include or address every consideration a rating committee may assess in assigning ratings to companies in this sector.

Exhibit 2

Oilfield services scorecard

	Weight	Aaa	Aa	A	Baa	Ba	B	Caa	Ca
Factor: Scale (20%)									
Revenue[1] (USD Billion)	10%	≥ $40	$20 - $40	$10 - $20	$4 - $10	$1.5 - $4	$0.5 - $1.5	$0.25 - $0.5	< $0.25
Total Assets[2] (USD Billion)	10%	≥ $100	$50 - $100	$20 - $50	$8 - $20	$4 - $8	$2 - $4	$0.5 - $2	< $0.5
Factor: Business Profile (30%)									
Business Profile	30%	Extremely stable EBITDA through the industry cycle; and global industry leader with operations in all major oil-and-gas producing countries; and provides an extremely diverse range of products and services to an extremely wide range of customers; and operates latest generation equipment or has consistently demonstrated superior technological capabilities that result in unparalleled and sustainable barriers to entry; and very long track record of exceptional operational performance; and exceptionally strong contractual revenue protections.	Very stable EBITDA through the industry cycle; and global industry leader with operations in numerous countries and in all major oil-and-gas producing regions; and provides a very diverse range of products and services to a very wide range of customers; and operates latest-generation equipment or has consistently demonstrated superior technological capabilities that result in extremely high barriers to entry; and long track record of exceptional operational performance; and very significant contractual revenue protections.	Stable EBITDA through the industry cycle; global competitor with operations in multiple countries and most major oil-and-gas producing regions; provides a diverse range of products and services to a wide range of customers; operates mostly latest-generation equipment or has mostly demonstrated leading-edge technological capabilities that result in very high barriers to entry; long track record of very strong operational performance; significant contractual revenue protections.	Modest EBITDA volatility through the industry cycle; global competitor with operations in a few countries and several major oil-and-gas producing regions; provides a diverse range of products and services to a moderately diversified customer base; operates mostly latest-generation equipment or has demonstrated strong technological capabilities that result in high barriers to entry; long track record of above-average operational performance; moderate contractual revenue protections.	High EBITDA volatility through the industry cycle; regionally diversified within a major oil-and-gas producing country; provides multiple products and services to a somewhat concentrated customer base; operates new and older equipment or has average technological capabilities, resulting in some barriers to entry; somewhat limited operating track record; some contractual revenue protections.	Very high EBITDA volatility through the industry cycle; operations in a limited number of regional markets within a country; provides a small number of products and services to a small number of customers; operates older equipment or has average or weaker-than-average technological capabilities, resulting in low barriers to entry; limited operating track record; very limited contractual revenue protections.	Very high EBITDA volatility through the industry cycle; operations in a single regional market; relies on a single product or service to a few customers; operates older equipment that is near the end of its useful life or has dated or unproven technological capabilities; short and inconsistent operating track record; no contractual revenue protections.	Extremely high EBITDA volatility through the industry cycle; or operations in a single small regional market; or relies on a single, niche product or service to one or two customers; or operates equipment that is in imminent need of replacement or has no technological capabilities; or weak or no operating track record.
Factor: Profitability and Efficiency (10%)									
EBIT Margin[3]	10%	≥ 40%	30% - 40%	20% - 30%	15% - 20%	10% - 15%	5% - 10%	2.5% - 5%	< 2.5%
Factor: Leverage and Coverage (20%)									
Debt / EBITDA[4]	10%	≤ 0.5x	0.5x - 1x	1x - 2x	2x - 3x	3x - 4x	4x - 6x	6x - 8x	> 8x
EBITDA / Interest Expense[5]	10%	≥ 30x	20x - 30x	10x - 20x	7x - 10x	4x - 7x	2x - 4x	1x - 2x	< 1x

	Weight	Aaa	Aa	A	Baa	Ba	B	Caa	Ca
Factor: Financial Policy (20%)									
Financial Policy	20%	Expected to have extremely conservative financial policies (including risk and liquidity management); very stable metrics; essentially no event risk that would cause a rating transition; and public commitment to a very strong credit profile over the long term.	Expected to have very conservative financial policies (including risk and liquidity management); stable metrics; minimal event risk that would cause a rating transition; and public commitment to a strong credit profile over the long term.	Expected to have predictable financial policies (including risk and liquidity management) that preserve creditor interests; although modest event risk exists, the effect on leverage is likely to be small and temporary; strong commitment to a solid credit profile.	Expected to have financial policies (including risk and liquidity management) that balance the interests of creditors and shareholders; some risk that debt-funded acquisitions or shareholder distributions could lead to a weaker credit profile.	Expected to have financial policies (including risk and liquidity management) that tend to favor shareholders over creditors; above-average financial risk resulting from shareholder distributions, acquisitions, or other significant capital structure changes.	Expected to have financial policies (including risk and liquidity management) that favor shareholders over creditors; high financial risk resulting from shareholder distributions, acquisitions, or other significant capital structure changes.	Expected to have financial policies (including risk and liquidity management) that create elevated risk of debt restructuring in varied economic environments.	Expected to have financial policies (including risk and liquidity management) that create elevated risk of debt restructuring even in healthy economic environments.

[1] For the linear scoring scale, the Aaa endpoint value is $60 billion. A value of $60 billion or better equates to a numeric score of 0.5. The Ca endpoint value is zero. A value of zero equates to a numeric score of 20.5.
[2] For the linear scoring scale, the Aaa endpoint value is $150 billion. A value of $150 billion or better equates to a numeric score of 0.5. The Ca endpoint value is zero. A value of zero equates to a numeric score of 20.5.
[3] For the linear scoring scale, the Aaa endpoint value is 70%. A value of 70% or better equates to a numeric score of 0.5. The Ca endpoint value is 0%. A value of 0% or worse equates to a numeric score of 20.5.
[4] For the linear scoring scale, the Aaa endpoint value is 0x. A value of 0x equates to a numeric score of 0.5. The Ca endpoint value is 12x. A value of 12x or worse equates to a numeric score of 20.5, as does a negative Debt/EBITDA value.
[5] For the linear scoring scale, the Aaa endpoint value is 40x. A value of 40x or better equates to a numeric score of 0.5. The Ca endpoint value is 0x. A value of 0x or worse equates to a numeric score of 20.5.
Source: Moody's Ratings

Discussion of the scorecard factors

In this section, we explain why each scorecard factor or subfactor is a meaningful credit indicator, and we discuss our analytical approach to scoring each factor or subfactor. When a factor comprises subfactors, we score at the subfactor level. Some factors do not have subfactors, in which case we score at the factor level.

In assessing quantitative subfactors, we use information that we calculate or estimate from a company's financial statements or regulatory filings, or that we derive from other observations. We may incorporate nonpublic information qualitatively in our assessment.

Our ratings are forward-looking and reflect our expectations for future financial and operating performance. However, historical results are helpful in understanding patterns and trends in a company's performance, as well as for peer comparisons. We typically calculate or estimate financial metrics, unless otherwise indicated, based on an annual or other 12-month period. However, we may assess the scorecard factors using various periods. For example, we may find it analytically useful to examine both historical and expected future performance for periods of several years or more.

All of the quantitative credit metrics in the scorecard incorporate our standard adjustments to income statement, cash flow statement and balance sheet amounts for items described in our cross-sector methodology on financial statement adjustments for nonfinancial corporations.[1] We may also make nonstandard adjustments that are specific to a particular company. See the appendix for examples of nonstandard adjustments that apply to this sector.

For a discussion of scorecard mechanics, see the "Scorecard-indicated outcome" section.

Factor: Scale (20% weight)

Why it matters

Scale is an important indicator of the overall strength of a company's business, the size of its market and the company's relevance within an industry and the economy. Scale is also an indicator of a company's importance to its customers and its bargaining strength with suppliers, as well as its ability to invest resources to maintain its competitive position and expand its marketing and geographic reach.

Companies with larger scale generally have more flexibility to manage their businesses under different demand and cost scenarios, an important consideration in an industry that is highly cyclical. They also tend to have greater access to capital markets, which provides the ability to adapt to competitive forces by undertaking major capital investment projects.

How we assess it for the scorecard

We score this factor based on two quantitative subfactors: Revenue and Total Assets.

Revenue

We use total reported revenue in billions of US dollars to measure scale, or to estimate it in the case of forward-looking expectations.

Total Assets

We use total assets for the latest fiscal year to measure scale, or to estimate it in the case of forward-looking expectations.

Factor: Business Profile (30% weight)

Why it matters

The business profile of a company is important because it greatly influences its ability to generate sustainable earnings and operating cash flow.

An oilfield services company with a stronger business profile has a leading and sustainable market position, is less vulnerable to fluctuating industry conditions in a particular region, has greater earnings consistency and predictability, and has less exposure to cash flow volatility. Core aspects of an oilfield services company's business profile are its earnings stability; geographic, product and customer diversity; market position; equipment and technological capability; operating track record and contractual revenue protections.

The consistency and predictability of an oilfield services company's earnings is an important indicator of its ability to maintain debt service payments. Oilfield services companies' earnings can be volatile, reflecting industry and macroeconomic cycles. Exploration and production (E&P) companies hire oilfield services companies to support the discovery, development and production of oil and gas resources. E&P companies typically manage reduced cash flows during downturns in hydrocarbon demand by decreasing capital spending on new projects and applying downward pressure on oilfield services companies' pricing, which can have a dramatic effect on the earnings and cash flows of oilfield services companies.

Oilfield services companies whose activities are predominantly related to oil and gas production tend to experience more stable demand than companies involved in exploration, drilling and completion activities, because production activities support current cash flow generation. On the other hand, exploration, drilling and completion activities involve significant cash outflows that E&P companies may suspend during downturns.

The geographic and product diversity of an oilfield services company and its market position are important because the ability to provide a diverse range of products and services to a wide range of customers expands revenue sources and boosts pricing power, which help maintain a company's cash flow through industry downturns. Companies that have market leadership positions or protected niches often have demonstrated the ability to provide best-in-class products and services and to attract and retain customers. The biggest and most diversified oilfield services companies are better able to provide products and services to large, multi-national E&P customers as well as national oil companies, which typically have large and stable capital expenditure budgets. Conversely, companies that provide standard or undifferentiated products and services and that have a limited geographic footprint are likely to experience greater competition and cash flow volatility.

The diversity of an oilfield services company's operations, products, services and customers is an important credit consideration. Diversification can reduce the impact of shifts in local market supply and demand, regulatory changes, competitors' actions, weather-related events and other market developments that could cause earnings erosion or volatility. A strong market position also allows an oilfield services company to hire superior talent and to negotiate better terms with suppliers and customers.

Companies with greater diversity and stronger market positions also have increased flexibility to adjust their geographic focus or product offerings in response to changes in market demand and economic and regulatory environments. Conversely, concentration in a limited number of geographic areas, product types and customers can expose an oilfield services company to risks related to changes in those markets and customers.

Equipment and technological capability are important because oilfield services companies that invest in and use the latest-generation equipment or have leading-edge technological capabilities often have a significant competitive advantage over competitors that employ older equipment or dated technology. Many oilfield services offerings are commodity-like, and product and service differentiation can be difficult to attain because competition is largely based on price (e.g., hydraulic fracturing services in North America). On the other hand, some oilfield services involve technological capabilities that can differentiate a service provider from its peers and are key to E&P customers reaching their hydrocarbon development objectives. For example, providers of drilling rigs have invested in rigs and automation designed for drilling in US shale basins to minimize drilling times, automate repeatable processes and accommodate pad drilling, resulting in safer operations, quicker drilling times and lower well-drilling costs.

An oilfield services company with technological advantages can benefit from reduced competition that results from high barriers to entry, and it can compete for business by offering new products and expanding to new markets. Additionally, technology leadership can support higher and more stable profit margins. Oilfield services companies that fail to keep their equipment and technological capability current risk losing business as their competitive position deteriorates.

The operating track record of an oilfield services company can provide important indications of its ability to deliver consistent and strong performance, which drives earnings and cash flows.

Customer contracts that include commitments for future equipment purchases (i.e., a backlog) or equipment leases are important because they provide an oilfield services company greater clarity into future revenue and can mitigate exposure to volatility in demand. Contracts for future services can provide some clarity on future revenue, but such contracts have typically extended for three years or less in the oilfield services sector in North America. Customers in some international regions enter into longer contracts.

How we assess it for the scorecard

We score this factor based on our qualitative assessment of the durability of demand and a company's ability to meet that demand.

We generally do not expect a given company to match each attribute listed for a given scoring category for this factor. We typically assign a factor score based on the alpha category for which a company has the greatest number of characteristics. However, there may be cases in which one characteristic is sufficiently important to a particular company's credit profile that it has a large influence on the factor score.

In our overall assessment of an oilfield services company's business profile, we consider its earnings stability, geographic, product and customer diversity, market position, equipment and technological capability, operating track record and contractual revenue protections.

Companies with greater earnings stability, stronger market positions, operations in several major oil-and-gas producing countries, product line and customer diversity, a fleet of the latest-generation equipment, leading-edge technological capabilities and contractual revenue protections typically receive higher scores for this factor.

Companies that provide services related to the production of oil and gas generally receive higher scores than companies that rely on services related to exploration and drilling, which can be subject to oil producers' capital spending cuts during downturns in commodity prices.

Factor: Profitability and Efficiency (10% weight)

Why it matters

A company's profitability matters because it supports the generation of sustainable cash flow and reflects the strength of the company's competitive position. Profitability also provides insights into a company's ability to manage key costs and pass them on to customers, which is an important indicator of the long-term viability of its business model. Strong profitability and operating efficiency can help a company to withstand economic and cyclical downturns that affect its competitive position and credit metrics.

Most oilfield services companies have large investments in fixed assets that are necessary to maintain and grow market share. The fixed assets require ongoing maintenance spending and need to be replaced or upgraded on a regular basis to invest in new technology and to offer a competitive service. However, high fixed costs cause greater margin volatility in a cyclical industry. Companies with greater profitability and capital efficiency are better positioned to maintain investment levels throughout the industry cycle.

How we assess it for the scorecard

We score this factor based on one quantitative subfactor: EBIT Margin.[2]

EBIT Margin

We use the ratio of earnings before interest and taxes (EBIT) to revenue.

Factor: Leverage and Coverage (20% weight)

Why it matters

Leverage and debt service coverage provide insight into the level of risk a company is willing to undertake. The degree of leverage is an important indicator of a company's financial flexibility, its ability to invest in growth opportunities and its capacity to pay debt service.

How we assess it for the scorecard

We score this factor based on two quantitative subfactors: Debt/EBITDA and EBITDA/Interest Expense.[3]

Debt / EBITDA

We use the ratio of total debt to earnings before interest, taxes, depreciation and amortization (EBITDA).

EBITDA / Interest Expense

We use the ratio of earnings before interest, taxes, depreciation and amortization (EBITDA) to interest expense.

Factor: Financial Policy (20% weight)

Why it matters

Financial policy is an important rating determinant because it directly affects a company's debt levels, its strategic direction, the risk of adverse changes in financing and capital structure, and its overall credit quality. Financial policy encompasses the tolerance of management and the board for financial risk and their level of commitment to maintaining a strong credit profile.

Financial risk tolerance serves as a guidepost for investment and capital allocation. Liquidity management is an important aspect of financial policy and can provide insight into risk tolerance. Liquidity management is distinct from the level of liquidity, which we discuss in the "Other considerations" section.

Many companies in this sector have historically used acquisitions to spur revenue growth, expand capabilities and product offerings, consolidate market positions, and replace and upgrade equipment. Acquisitions can have an important impact on an oilfield services company's credit profile, depending in part on the existing capital structure and the extent to which it is changed by the transaction. In addition, oilfield services companies' decisions regarding shareholder distributions compared with investment in equipment provide insight into financial policies. Oilfield services companies have encountered requests for increased shareholder distributions during periods of peak profitability, potentially at the expense of investment in offshore drilling rigs and pressure pumping equipment.

How we assess it for the scorecard

We assess qualitatively a company's targeted credit profile or desired capital structure, and its ability to achieve its targets. We also consider a company's complexity and transparency, its track record of risk and liquidity management, and its public commitments and history of adhering to its commitments in these areas. We also consider management's operating performance and use of cash flow through different phases of economic and industry cycles, as well as how management responds to key events, such as changes in the credit markets and liquidity environment, legal actions, competitive challenges and regulatory pressures.

When considering event risks in the context of scoring financial policy, we assess the likelihood and potential effect of M&A or other types of balance-sheet-transforming events. We assess management's appetite for M&A activity, with a focus on the type of transactions (i.e., core competency or new business) and funding decisions. We also evaluate the frequency and materiality of acquisitions and previous financing choices. A history of debt-financed or credit-transforming acquisitions generally results in a lower score for this factor. We may also consider a company's willingness to sell itself and the negative repercussions that could result, such as through a leveraged buyout transaction.

We also consider a company's and its owners' past record of balancing shareholder returns and debtholders' interests. We are likely to take a negative view in scoring this factor when a company has a track record of favoring shareholder returns at the expense of debtholders.

Other considerations

Ratings may reflect other considerations that are not in the scorecard, usually because the credit importance of these considerations varies widely among companies in this sector or because they may be important only under certain circumstances or for a subset of companies. Such considerations can include financial controls and the quality of financial reporting; corporate legal structure; the quality and experience of management; assessments of corporate governance, as well as environmental and social considerations; exposure to uncertain licensing regimes; and possible government interference in some countries. Regulatory, litigation, liquidity, technology and reputational risk, as well as changes to consumer and business spending patterns, competitor strategies and macroeconomic trends can also be considerations that affect ratings.

The following are examples of other considerations that may be reflected in our ratings and may cause ratings to be different from scorecard-indicated outcomes.

Environmental, social and governance considerations

Where environmental, social and governance (ESG) issues are meaningful for credit profiles, we incorporate them into our ratings analysis in a variety of ways in the application of our sector-specific methodologies. As one part of our overall credit analysis, we consider how ESG considerations could affect the qualitative and quantitative factors and subfactors in the scorecard.

Even where ESG considerations do not affect the measures in the scorecard, or where they cannot be quantified, we incorporate them into our overall analysis of credit drivers that are meaningful to the rating. As a result, we may incorporate these ESG risks qualitatively outside of the scorecard. As part of our ratings analysis, we may establish Issuer Profile Scores (IPSs), which indicate our opinion of the extent to which a given company is exposed to E, S and G risks (incorporating ESG-specific mitigants) or benefits from its exposure to E, S or G. The IPSs are inputs to credit ratings. For more information, see our methodology that describes our general principles for assessing ESG risks.[4]

Companies in the oil and gas sector are exposed to significant risks from the global effort to curb greenhouse gas emissions. Oilfield services companies generally face environmental risks similar to those of the E&P companies that are their customers. E&P companies hire oilfield services companies to discover, develop and produce oil and natural gas resources. A dampening of oil demand due to the increased adoption of alternative energy sources would reduce service demand and weaken the pricing power of oilfield services companies.

Stricter environmental policies and regulations can also lead to higher costs for oilfield services companies. The severity and immediacy of these risks depends on the extent of the carbon emissions curbs and the timing of their implementation. In addition, the specific mechanisms used to curb carbon emissions may affect individual regions and companies differently. Oilfield services companies have historically been able to pass on incremental environmental or regulatory costs to customers.

Demographic and societal trend risk for oilfield services companies is closely linked to carbon transition risk. The increasing social awareness of climate change and negative views of traditional fossil fuel industries have triggered policy shifts that encourage investment in renewable energy sources and accelerate the decarbonization process. Earnings from existing projects may be affected where policy changes or company decarbonization efforts shorten the expected life or profitability of such projects.

Oilfield services companies are generally indemnified by their E&P customers for most drilling and other services-related activity, and particularly for high-risk endeavors, thereby minimizing their exposure to potential legal claims or regulatory penalties. Nevertheless, a catastrophic oil spill or similar event could expose an oilfield services company to substantial financial liability. At a minimum, such events could temporarily impair an oilfield services company's ability to operate, tarnish its reputation and have a negative effect on its credit profile.

The long-term nature of carbon transition risks may mean that they are not fully reflected in our published scorecards. Forward-looking inputs to our published scorecards are typically based on our near-term projections, in part because we may not have sufficient visibility into a company's future results beyond this horizon that would enable us to accurately incorporate these risks in assessing scorecard factors and subfactors, especially quantitative factors. Like other long-term risks that may not be captured in a near-term forward view, carbon transition risks may be incorporated qualitatively outside the scorecard, based on our view of trends that extend beyond the horizon for which more precise projections are practicable. As a result, carbon transition risks may, over time, cause our ratings to be lower than scorecard-indicated outcomes for some companies in this sector.

Among the areas of focus in corporate governance are risk management, audit committee financial expertise, the incentives created by executive compensation packages, related-party transactions, interactions with outside auditors and ownership structure.

Liquidity

Liquidity is an important rating consideration for all companies, and extremely weak liquidity can heavily affect ratings in many cases. Our liquidity assessment can be particularly important where there are significant short-term demands on a company's liquidity, for example, in highly seasonal operating environments where working capital needs must be considered. We form an opinion on likely short-term liquidity requirements from the perspective of both sources and uses of cash. For more information, see our liquidity cross-sector methodology.[5]

Cyclical sectors

In a cyclical sector, the scorecard-indicated outcome may be higher than the assigned rating at the top of the economic cycle and lower than the assigned rating at the bottom of the cycle. While using annual or other 12-month-period financials in the scorecard typically provides useful insights into recent or near-term financial results, ratings may also reflect our expectations for the progression of financial results over a longer period that may include a full economic cycle. However, cyclicality itself poses many different types of

risks to companies, and cycles do not reverse themselves with predictable regularity. A secular decline or expansion may also affect a cyclical sector. Ratings may incorporate these considerations qualitatively.

Management strategy

The quality of management is an important factor supporting a company's credit strength. Assessing the execution of business plans over time can be helpful in assessing management's business strategies, policies and philosophies, and in evaluating management performance relative to that of competitors and our projections. Management's track record of adhering to stated plans, commitments and guidelines provides insight into management's likely future performance, including in stressed situations.

Financial controls

We rely on the accuracy of audited financial statements to assign and monitor ratings in this sector. The quality of financial statements may be influenced by internal controls, including the proper tone at the top, centralized operations, and consistency in accounting policies and procedures. Auditors' reports on the effectiveness of internal controls, auditors' comments in financial reports and unusual restatements of financial statements or delays in regulatory filings may indicate weaknesses in internal controls.

Excess cash balances

Some companies in this sector may maintain cash balances (meaning liquid short-term investments, as well as cash) that far exceed their operating needs. This excess cash can be an important credit consideration.

The amount of cash required to run a business can vary materially based on the regions of operation and the specific subsectors in which a company operates. As a result, we typically assess cash on the balance sheet qualitatively, taking into account a company's track record and its financial and liquidity policies. Cash holdings may be a material mitigant in our scenario analyses of potential acquisitions, share buybacks or special dividends. We assess possible uses of cash, especially for companies that may be acquisitive or under pressure to return cash to shareholders. By contrast, some companies maintain large cash holdings for long periods in excess of their operating and liquidity needs solely because of conservative financial policies, which provides a stronger indication of an enduring approach that will benefit creditors. Large cash holdings provide some buffers against credit deterioration, and we typically consider cash balances in our analysis of near-term liquidity sources and uses. See the "Liquidity" section for more information.

In accordance with our view of how a company will use its cash, we may use total debt or net debt in our forward-looking view of credit metrics for the Leverage and Coverage factor. We may also cite rating thresholds based on these metrics.

Event risk

We also recognize the possibility that an unexpected event could cause a sudden and sharp decline in a company's fundamental creditworthiness, which may cause the assigned rating to be lower than the scorecard-indicated outcome. Event risks — which vary and can range from leveraged recapitalizations to sudden regulatory changes or liabilities from an accident — can overwhelm even a stable, well-capitalized company. Other types of event risks may include M&A, asset sales, spinoffs, shareholder distributions, litigation, natural disasters, pandemics, significant cybercrime events and geopolitical conflicts.

Regulatory considerations

Companies in this sector are subject to varying degrees of regulatory oversight. Effects of these regulations may entail limitations on operations, higher costs, and higher potential for technology disruptions and demand substitution. Regional differences in regulation, implementation or enforcement may advantage or disadvantage a particular company.

Our view of future regulatory requirements is important because it is the basis for our expectations of future financial metrics and of a company's ability to generate sufficient cash flow relative to its debt burden over the medium and longer term.

Parental support

Ownership can provide ratings lift for a particular company if it is owned by a highly rated owner or owners who view it to be of strategic importance. In our analysis of parental support, we consider whether the parent has the financial capacity and strategic incentives to provide support to the company in times of stress or financial need (e.g., a major capital investment or advantaged operating agreement), or if it has done so in the past. Conversely, if the parent puts a high dividend burden on the company, which in turn reduces its flexibility, the ratings will reflect this risk. For more information, see our cross-sector methodology on assessing the impact of financial support from a parent, subsidiary or affiliate in the absence of a guarantee.[6]

Government-related issuers may receive ratings uplift from expected government support. However, for certain companies, government ownership can have a negative effect on the underlying Baseline Credit Assessment (BCA). For example, price controls, onerous taxation and high distributions can have a negative effect on a company's underlying credit profile. For more information on the BCA, see *Rating Symbols and Definitions* and our cross-sector methodology for government-related issuers.[7]

Additional metrics

The metrics included in the scorecard are those that are generally most important in assigning ratings to companies in this sector; however, we may use additional metrics to inform our analysis of specific companies. These additional metrics may be important in our forward-looking view of metrics in the scorecard or in other rating considerations.

Non-wholly owned subsidiaries

Some companies in this sector choose to dilute their equity stake in certain material subsidiaries, such as through an initial public offering or a sale to a private investor, which may, in some cases, negatively affect future financial flexibility. While such transactions may improve cash holdings on a one-off basis, selling minority interests in subsidiaries may have a negative effect on cash flow available to the parent company that consolidated financial statements may not fully reflect. For example, in the case of an equity stake reduction in a subsidiary down to 75%, in the parent's financial statements, the subsidiary's revenue and EBITDA would typically still be consolidated at the group level.

The presence of minority stakes can increase structural subordination and reduce the parent's share of dividend flows from a non-wholly owned subsidiary. While less frequent, the sale of a minority stake may have provisions that further limit the parent's financial flexibility, for instance restrictions on cash pooling with other members of the corporate family, limitations on dividends and distributions, or arms-length business requirements. Minority stakeholders may have seats on the board of the subsidiary.

In many cases, we consider the impact of non-wholly owned subsidiaries qualitatively. For example, we may find that an additional view of financial results, such as analyzing cash flow on a proportional consolidation basis, may be useful to inform our analysis. When equity dilution or structural subordination from non-wholly owned subsidiaries is material and negative, the credit impact is captured in our ratings.

For companies that hold material minority interest stakes (investor companies), consolidated funds from operations typically includes the dividends received from the equity-accounted entity, while none of its debt is consolidated. The investor company's cash flow may be subject to interruption if it is required for the equity-accounted entity's debt service, capital expenditures or other cash needs. The debt of the equity-accounted entity is structurally senior to the investor company's debt. When an equity-accounted entity has a material level of debt, we may estimate one or more of the investor company's financial metrics using proportional consolidation or another approach to inform our analysis of consolidated financials and capture the credit impact of these considerations in our ratings.

Other institutional support

In some countries, large companies have received government or banking support in the event of financial difficulties because of their overall importance to the functioning of the economy. We consider whether group and banking relationships may provide support when systemically important companies encounter significant financial stress.

As a notable example, some of our corporate ratings in Japan incorporate the government or banking support we expect to be provided, based on historical experience, to large and systemically important companies. Over the years, this support has resulted in lower levels of default for Japanese companies than might otherwise have occurred.

Scorecard-indicated outcome

We arrive at a scorecard-indicated outcome by first mapping the outcome for each factor or subfactor to a numeric score, and then by producing an aggregate numeric score and the equivalent alphanumeric outcome.

Mapping scorecard factors to a numeric score

We map the outcome of each factor or subfactor in the scorecard to a broad Moody's rating category and to a corresponding numeric score. We score qualitative factors and subfactors based on the description by broad rating category in the scorecard. The following exhibit shows the numeric score that corresponds to each broad rating category.

Exhibit 3
Numeric scores for qualitative factors and subfactors

Aaa	Aa	A	Baa	Ba	B	Caa	Ca
1	3	6	9	12	15	18	20

Source: Moody's Ratings

We score quantitative factors and subfactors on a linear continuum. For each metric, the scorecard shows the range by broad rating category. We use the scale in the following exhibit and linear interpolation to convert the metric, based on its placement within the scorecard range, into a numeric score, which may be a fraction. As an illustrative example, if the Baa range for the ratio of revenue to interest is 50x to 100x, the numeric score for a company with a ratio of 99x – relatively strong within this range – would be closer to 7.5, while the numeric score for a company with a ratio of 51x – relatively weak within this range – would be closer to 10.5. In the scorecard footnotes, we define the endpoints of the line (i.e., the metric values that equate to the highest possible numeric score and the lowest possible numeric score).

Exhibit 4
Ranges of numeric scores for quantitative factors and subfactors

Aaa	Aa	A	Baa	Ba	B	Caa	Ca
0.5 - 1.5	1.5 - 4.5	4.5 - 7.5	7.5 - 10.5	10.5 - 13.5	13.5 - 16.5	16.5 - 19.5	19.5 - 20.5

Source: Moody's Ratings

Determining the overall scorecard-indicated outcome

We multiply the numeric score for each subfactor (or each factor, when the factor has no subfactors) by the weight of that subfactor (or factor) and sum the results to produce an aggregate numeric score. We map the aggregate numeric score back to a scorecard-indicated outcome using the ranges in the following exhibit.

Exhibit 5
Scorecard-indicated outcome

Aggregate numeric score	Scorecard-indicated outcome
x ≤ 1.5	Aaa
1.5 < x ≤ 2.5	Aa1
2.5 < x ≤ 3.5	Aa2
3.5 < x ≤ 4.5	Aa3
4.5 < x ≤ 5.5	A1
5.5 < x ≤ 6.5	A2
6.5 < x ≤ 7.5	A3
7.5 < x ≤ 8.5	Baa1
8.5 < x ≤ 9.5	Baa2
9.5 < x ≤ 10.5	Baa3
10.5 < x ≤ 11.5	Ba1
11.5 < x ≤ 12.5	Ba2
12.5 < x ≤ 13.5	Ba3
13.5 < x ≤ 14.5	B1
14.5 < x ≤ 15.5	B2
15.5 < x ≤ 16.5	B3
16.5 < x ≤ 17.5	Caa1
17.5 < x ≤ 18.5	Caa2
18.5 < x ≤ 19.5	Caa3
19.5 < x ≤ 20.5	Ca
x > 20.5	C

Source: Moody's Ratings

For example, a company with an aggregate numeric score of 11.7 would have a Ba2 scorecard-indicated outcome.

In general, the scorecard-indicated outcome is oriented to the senior unsecured rating for investment-grade companies and the corporate family rating (CFR) for speculative-grade companies. For companies that benefit from rating uplift from parental support, government ownership or other institutional support, we consider the underlying credit strength or BCA for comparison to the scorecard-indicated outcome. For more information on the BCA, see *Rating Symbols and Definitions* and our cross-sector methodology for government-related issuers.[8]

Assigning issuer-level and instrument-level ratings

After considering the scorecard-indicated outcome, other considerations and relevant cross-sector methodologies, we typically assign a senior unsecured rating for investment-grade companies and a CFR for speculative-grade companies. For government-related issuers, we assign a BCA.

We may notch individual debt instrument ratings upward or downward from the senior unsecured rating or the CFR to reflect our assessment of differences in expected loss related to an instrument's seniority level and collateral. For more information, see our cross-sector methodologies on loss given default for speculative-grade nonfinancial companies, notching corporate instrument ratings based on differences in security and priority of claim, and assigning short-term ratings.[9]

Key rating assumptions

For information about key rating assumptions that apply to methodologies generally, see *Rating Symbols and Definitions*.[10]

Limitations

In the preceding sections, we have discussed the scorecard factors and other considerations that may be important in assigning ratings. In this section, we discuss limitations that pertain to the scorecard and to the overall methodology.

Limitations of the scorecard

There are various reasons why scorecard-indicated outcomes may not map closely to assigned ratings.

For example, the scorecard may not fully capture credit loss and recovery considerations, which typically become more important as a company gets closer to default. The scorecard is also limited by its upper and lower bounds, causing scorecard-indicated outcomes to be less likely to align with ratings for companies at the upper and lower ends of the rating scale.

The weights of each scorecard factor and subfactor approximate their importance for rating decisions across the sector, but the importance of a particular factor may vary substantially based on an individual company's circumstances.

Factors outside the scorecard, including those discussed in the "Other considerations" section, may be important for ratings, and their relative importance may also vary from company to company. In addition, certain broad methodological considerations described in one or more cross-sector methodologies may be relevant to ratings in this sector.[11] These considerations include how sovereign credit quality affects nonsovereign issuers, the assessment of credit support from other entities, the relative ranking of different classes of debt and hybrid securities, and the assignment of short-term ratings.

We may use the scorecard over various historical or forward-looking periods. Furthermore, in our ratings we often incorporate directional views of risks and mitigants in a qualitative way.

General limitations of the methodology

This methodology does not include an exhaustive description of all the factors we may consider in assigning ratings in this sector. Companies in this sector may face new risks or new combinations of risks, and they may develop new strategies to mitigate risk. We seek to incorporate all material credit considerations in our ratings and to take the most forward-looking perspective that visibility into these risks and mitigants permits.

Ratings reflect our expectations for a company's future performance; however, as the forward horizon lengthens, uncertainty increases and the utility of precise estimates, as scorecard inputs or in other considerations, typically diminishes. Our forward-looking opinions are based on assumptions that may prove, in hindsight, to have been incorrect. Reasons for this could include unanticipated changes

in any of the following: the macroeconomic environment, general financial market conditions, industry competition, disruptive technology, or regulatory and legal actions. In any case, predicting the future is subject to substantial uncertainty.

Appendix: Nonstandard adjustments to financial statements

In addition to making standard adjustments to a company's financial statements, we may also make nonstandard adjustments to better reflect underlying economics, improve comparability with peer companies and reflect estimates or assumptions we consider to be more appropriate for credit analysis.

We make nonstandard financial statement adjustments based on public information, although we may consider private information in our ratings. We may apply a materiality threshold to determine whether or not to make a nonstandard adjustment to a particular company's financial statements.

In the oilfield services sector, we adjust certain metrics of companies that primarily provide seismic services to reflect our view that spending related to multi-client surveys is an operating cost that should be expensed when it is incurred. In multi-client surveys, a company develops a library of seismic data and seeks to sell that data to multiple prospective customers.

Nonstandard adjustments for these types of companies are as follows:

» adjust assets to remove the amount of multi-client survey expenditure that has been capitalized;

» reduce EBIT and EBITDA by the amount incurred in the period; and

» adjust EBIT to exclude amortization of capitalized multi-client expenditures.

Moody's related publications

Credit ratings are primarily determined through the application of sector methodologies. Certain broad methodological considerations (described in one or more cross-sector methodologies) may also be relevant to the determination of credit ratings of issuers and instruments. A list of sector and cross-sector methodologies can be found here.

Data summarizing the historical robustness and predictive power of credit ratings can be found here.

For more information, see *Rating Symbols and Definitions*, which is available here.

Endnotes

1 A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

2 For definitions of our most common metrics, see our cross-sector methodology that describes our financial statement adjustments for nonfinancial corporations. A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

3 For definitions of our most common metrics, see our cross-sector methodology that describes our financial statement adjustments for nonfinancial corporations. A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

4 A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

5 A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

6 A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

7 Links to *Rating Symbols and Definitions* and to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

8 Links to *Rating Symbols and Definitions* and to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

9 A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

10 A link to *Rating Symbols and Definitions* can be found in the "Moody's related publications" section.

11 A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

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